Intelligent Systems Coursework Report

Title: Recruitment Automation System Using Agentic AI

Student Name:

Student ID:

Table of Contents

1.0 Application and Workflow Selection....3

1.1 Introduction and Motivation of the Selected Application....4

1.2 Explanation of the Selected Workflow with Diagram....4

1.3 Objectives of the Selected Intelligent System....5

1.4 Impact of the Selected Project if Developed....5

2.0 Selection of AI Technologies....6

2.1 Identification of Similar Systems for Tool Selection....6

2.2 Innovative and Original Approaches....6

2.3 AI Methods, Technologies, and Justifications....6

3.0 Simulations & Development Setup....7

3.1 Simulation Environments and Setup Parameters....7

3.2 Agent-wise Task Distribution and Functionality....8

3.3 Encoding and Data Processing Prior to Simulation....9

4.0 Use Case Explanation with Evidence....10

4.1 Step-by-Step Workflow Execution....10

4.2 Developed User Interface and External Interaction....14

4.2.1 User Interface Design....14

4.2.2 External Interaction and Integration....15

5.0 Conclusions....15

5.1 Research Problem Addressed....15

5.2 Summary of Findings....15

Reference....17

List of Figures:

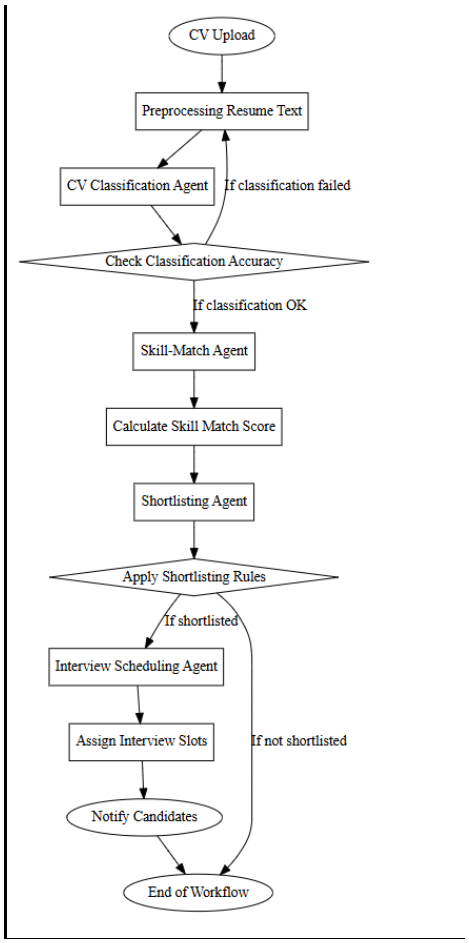
Figure 1 Work Flow Diagram 5
Figure 2 Simulation Environments and Setup Parameters 8
Figure 3 Encoding and Data Processing Prior to Simulation 10
Figure 4 CV Classification 11
Figure 5 Skill-Match Agent 12
Figure 6 Shortlisting Agent 13
Figure 7 Interview Scheduling Agent 14

1.0 Application and Workflow Selection

1.1 Introduction and Motivation of the Selected Application

With the current competitive job market, most organisations have been challenged in managing the recruitment process effectively, as the number of applications received is quite high. Resume screening is time-consuming, manual and a tedious procedure, and it may result in the loss of quite suitable candidates. An easy way of dealing with these inefficiencies is to create an intelligent system that breaks down the recruitment procedure by automating CV classification, skill assessment, shortlisting, and the scheduling of interviews. This system simulates human decision-making at several levels by applying agentic AI methods and has consistency, scalability, and accuracy(Joshi, 2025). The incentive to use this application is that it is practical in terms of its application to human resource management, the possibility of decreasing the operational costs, and the possibility of enhancing the experience of the candidates by simplifying the recruitment processes.

[redacted]



[redacted]

[redacted]

[redacted]

[redacted]

[redacted]

[redacted]

[redacted]

[redacted]

[redacted]

[redacted]

[redacted]

[redacted]

[redacted]

[redacted]

[redacted]

[redacted]

[redacted]

[redacted]

[redacted]

[redacted]

[redacted]

[redacted]

[redacted]

[redacted]

[redacted]

[redacted]

[redacted]

[redacted]

```python
# Install Kaggle
!pip install -q kaggle

KAGGLE_USERNAME = "your_kaggle_username"
KAGGLE_KEY = "your_kaggle_api_key"

import os

# Set environment variables
os.environ["KAGGLE_USERNAME"] = KAGGLE_USERNAME
os.environ["KAGGLE_KEY"] = KAGGLE_KEY

# Create directory for dataset
os.makedirs("resume_dataset", exist_ok=True)

# Download the dataset
!kaggle datasets download -d snehaanbhawal/resume-dataset -p resume_dataset

# Unzip the dataset
!unzip -o resume_dataset/*.zip -d resume_dataset

# List downloaded files
os.listdir("resume_dataset")
```

```
Dataset URL: https://www.kaggle.com/datasets/snehaanbhawal/resume-dataset
License(s): CC0-1.0
Downloading resume-dataset.zip to resume_dataset
  0% 0.00/62.5M [00:00<?, ?B/s]
100% 62.5M/62.5M [00:00<00:00, 1.06GB/s]
Archive:  resume_dataset/resume-dataset.zip
  inflating: resume_dataset/Resume/Resume.csv
  inflating: resume dataset/data/data/ACCOUNTANT/10554236.pdf
```

```python
# Step 1: Import libraries
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
from sklearn.model_selection import train_test_split
from sklearn.feature_extraction.text import TfidfVectorizer
from sklearn.ensemble import RandomForestClassifier
from sklearn.metrics import accuracy_score, classification_report, confusion_matrix
```

```python
# Step 1: Import libraries
import pandas as pd
import re
import matplotlib.pyplot as plt
import seaborn as sns
```

```python
# Step 1: Import libraries
import numpy as np
import datetime
```

[redacted]

[redacted]

[redacted]

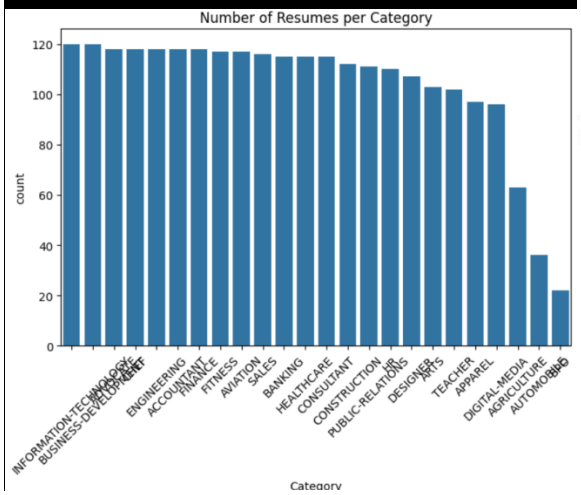
[redacted]

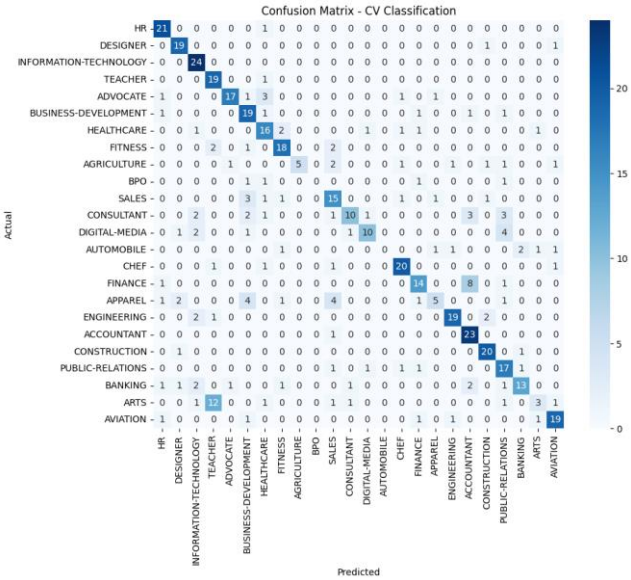
[redacted]

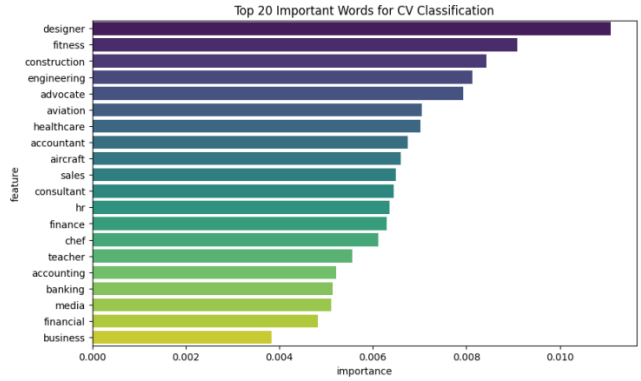
[redacted]

```python
# Step 2: Load dataset
df = pd.read_csv('/content/resume_dataset/Resume/Resume.csv')

# Step 3: Basic Visualization - Category distribution
plt.figure(figsize=(8,5))
sns.countplot(x='Category', data=df, order=df['Category'].value_counts().index)
plt.title("Number of Resumes per Category")
plt.xticks(rotation=45)
plt.show()

# Step 4: Preprocess text
df['Resume_str'] = df['Resume_str'].str.lower()  # Lowercase
df['Resume_str'] = df['Resume_str'].str.replace(r'\W', ' ', regex=True)  # Remove special characters
df['Resume_str'] = df['Resume_str'].str.replace(r'\s+', ' ', regex=True)  # Remove extra spaces

# Step 5: Split dataset
X = df['Resume_str']
y = df['Category']

X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42, stratify=y)

# Step 6: Feature extraction using TF-IDF
vectorizer = TfidfVectorizer(max_features=3000)
X_train_tfidf = vectorizer.fit_transform(X_train)
X_test_tfidf = vectorizer.transform(X_test)
```

[redacted]







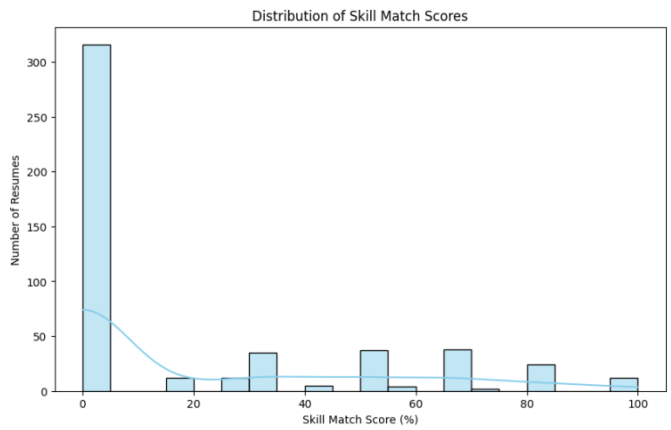

```
Accuracy: 0.6961770623742455

Classification Report:
                      precision    recall  f1-score   support

          ACCOUNTANT       0.62      0.96      0.75        24
            ADVOCATE       0.89      0.71      0.79        24
         AGRICULTURE       1.00      0.38      0.56        13
             APPAREL       0.62      0.26      0.37        19
                ARTS       0.50      0.14      0.22        21
          AUTOMOBILE       0.00      0.00      0.00         7
           AVIATION       0.79      0.79      0.79        24
            BANKING       0.76      0.57      0.65        23
                BPO       0.00      0.00      0.00         4
BUSINESS-DEVELOPMENT       0.58      0.79      0.67        24
               CHEF       0.80      0.83      0.82        24
       CONSTRUCTION       0.80      0.91      0.85        22
         CONSULTANT       0.77      0.43      0.56        23
           DESIGNER       0.79      0.90      0.84        21
      DIGITAL-MEDIA       0.77      0.53      0.62        19
        ENGINEERING       0.86      0.79      0.83        24
            FINANCE       0.70      0.58      0.64        24
            FITNESS       0.75      0.78      0.77        23
         HEALTHCARE       0.59      0.70      0.64        23
                 HR       0.78      0.95      0.86        22
INFORMATION-TECHNOLOGY       0.71      1.00      0.83        24
     PUBLIC-RELATIONS       0.55      0.77      0.64        22
              SALES       0.54      0.65      0.59        23
            TEACHER       0.54      0.95      0.69        20
```

[redacted]

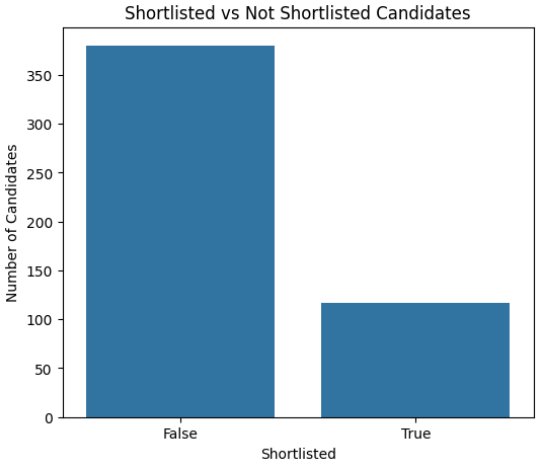
```
           accuracy                           0.70       497
          macro avg       0.66      0.64      0.62       497
       weighted avg       0.70      0.70      0.67       497

/usr/local/lib/python3.12/dist-packages/sklearn/metrics/_classification.py:1565: UndefinedMetricWarning: Preci
  _warn_prf(average, modifier, f"{metric.capitalize()} is", len(result))
/usr/local/lib/python3.12/dist-packages/sklearn/metrics/_classification.py:1565: UndefinedMetricWarning: Preci
  _warn_prf(average, modifier, f"{metric.capitalize()} is", len(result))
/usr/local/lib/python3.12/dist-packages/sklearn/metrics/_classification.py:1565: UndefinedMetricWarning: Preci
  _warn_prf(average, modifier, f"{metric.capitalize()} is", len(result))
```

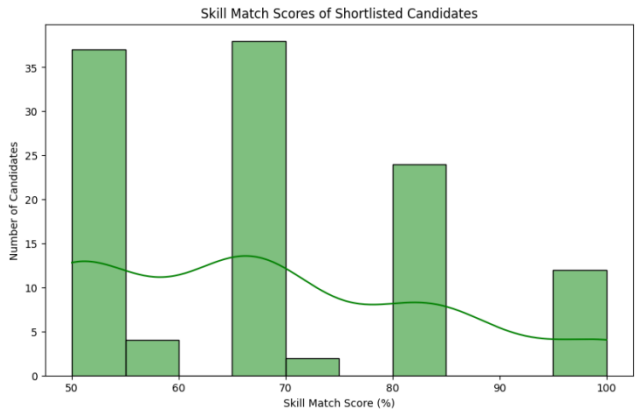
[redacted]

[redacted]



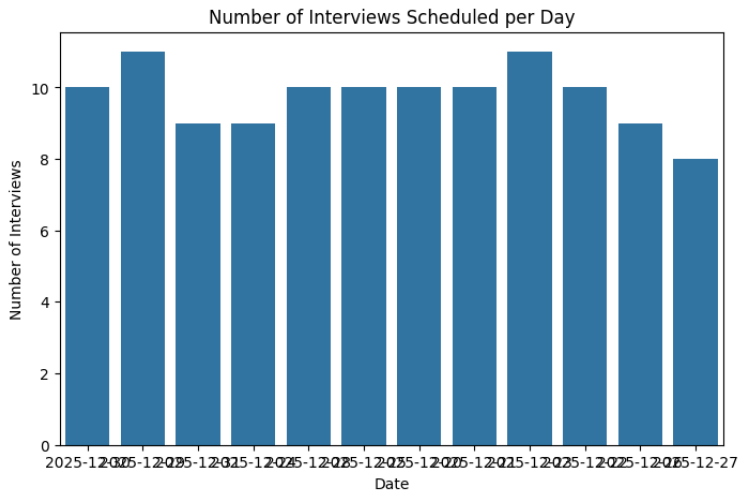

```
                                                Resume   Job Category  \
0   designer summary to get a strong foothold on ...       DESIGNER
1   digital marketing director summary with a bac...  DIGITAL-MEDIA
2   laboer floor construction worker all a round ...   CONSTRUCTION
3   medical record technician professional summar...     HEALTHCARE
4   construction manager project coordinator insp...   CONSTRUCTION
5   hr generalist experience 04 2015 to current p...             HR
6   claims service manger professional summary se...     AUTOMOBILE
7   videographer professional profile creative in...  DIGITAL-MEDIA
8   senior marketing automation consultant summar...           ARTS
9   digital strategy consultant summary to furthe...  DIGITAL-MEDIA

   Skill Match Score Matched Skills
0              16.67       [design]
1               0.00            []
2               0.00            []
3               0.00            []
4               0.00            []
5              16.67     [training]
6               0.00            []
7               0.00            []
8               0.00            []
9               0.00            []
```






```
Number of shortlisted candidates: 117
                                                Resume           Job Category  \
0   lighting field engineering technician summary...            ENGINEERING
1   finance manager finance manager executive pro...                FINANCE
2   objective to obtain a position in the informa...  INFORMATION-TECHNOLOGY
3   sales executive summary sales and customer se...                  SALES
4   student finance consultant summary customer s...                FINANCE
5   finance manager summary skilled finance leade...                FINANCE
6   manager engineering manufacturing systems eng...            ENGINEERING
7   preschool teacher summary enthusiastic presch...                TEACHER
8   engineering technician professional summary s...            ENGINEERING
9   sales associate skills teamwork problem solvi...                  SALES

   Skill Match Score                                Matched Skills  \
0             66.67             [engineering, design, project, cad]
1             83.33    [finance, budget, accounting, analysis, tax]
2             71.43  [python, java, network, software, programming]
3             66.67          [marketing, crm, sales, communication]
4             50.00                 [finance, accounting, analysis]
5             66.67         [finance, budget, accounting, analysis]
6             66.67     [engineering, design, project, development]
7             50.00                   [teaching, lesson, education]
8             66.67     [engineering, design, project, development]
9             50.00                [marketing, sales, communication]
...
6         True
7         True
8         True
9         True
Output is truncated. View as a scrollable element or open in a text editor. Adjust cell output settings...
```

[redacted]




```
                                                Resume            Job Category
0   lighting field engineering technician summary...             ENGINEERING
1   finance manager finance manager executive pro...                 FINANCE
2   objective to obtain a position in the informa...  INFORMATION-TECHNOLOGY
3   sales executive summary sales and customer se...                   SALES
4   student finance consultant summary customer s...                 FINANCE
5   finance manager summary skilled finance leade...                 FINANCE
6   manager engineering manufacturing systems eng...             ENGINEERING
7   preschool teacher summary enthusiastic presch...                 TEACHER
8   engineering technician professional summary s...             ENGINEERING
9   sales associate skills teamwork problem solvi...                   SALES

   Skill Match Score      Interview Slot
0             66.67 2025-12-30 09:00:00
1             83.33 2025-12-29 14:00:00
2             71.43 2025-12-29 10:00:00
3             66.67 2025-12-31 15:00:00
4             50.00 2025-12-24 10:00:00
5             66.67 2025-12-29 16:00:00
6             66.67 2025-12-28 14:00:00
7             50.00 2025-12-25 11:00:00
8             66.67 2025-12-20 09:00:00
9             50.00 2025-12-21 11:00:00
```

[redacted]

[redacted]

[redacted]

[redacted]

[redacted]

[redacted]

[redacted]

[redacted]

[redacted]

[redacted]

[redacted]

[redacted]

[redacted]

[redacted]

[redacted]

[redacted]

[redacted]

[redacted]

[redacted]

[redacted]

[redacted]

[redacted]

[redacted]

[redacted]

[redacted]

[redacted]

[redacted]

[redacted]

[redacted]

[redacted]

[redacted]